

Bird Species Observation Analysis

Forest and grassland point-count surveys
11 National Park Service units, 2018 breeding season

Dataset	15,372 sightings, 1,408 survey sessions, 126 species
Coverage	11 National Park Service units, May to July 2018
Habitats	Forest and grassland point-count surveys
Report generated	31 August 2026
Source	analysis.run_all() - every figure computed at build time

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Executive summary

This report analyses 15,372 bird sightings recorded across 1,408 point-count survey sessions in 11 National Park Service units during the 2018 breeding season, covering 126 species in forest and grassland habitats. Thirteen analytical questions were posed. **Five produced findings that survived scrutiny**, one was rejected outright once the comparison was made fair, and the remainder are descriptive or methodological.

The most consequential result is methodological rather than ecological. The difference between the three surveyors (2.69 species per session) is roughly 54 times larger than the habitat difference the survey was designed to measure (0.05). The study's conclusions survive only because observer assignment was balanced across habitats and parks.

Headline findings

- **Forest shelters Wood Thrush, not at-risk birds as a class.** At-risk species appear in 3.65% of forest sightings against 0.59% in grassland - a 6.2x gap ($p = 1.66e-18$) holding in all 4 shared parks. However 82% of those sightings are a single species; excluding it the gap falls to 2.1x and survives in only 2 of 4 parks.
- **Grassland supports habitat specialists; forest supports none.** Of 48 well-sampled species, 17 are grassland specialists and 0 are forest specialists.
- **Disturbance is the largest environmental effect measured.** Serious disturbance reduces recorded species by 36.8% (8.93 to 5.65 per session, $p = 2.52e-15$).
- **Grassland surveys are most productive early.** Richness falls 17.6% between the 5-6am and 9-10am bands ($p = 1.75e-05$), confirmed by an independent whole-morning trend test. Forest shows no such effect.
- **Observer identity outweighs every ecological variable.** A 37.0% spread between three surveyors, significant in every pairwise comparison and consistent in every park.
- **The habitats share a species pool but weight it differently.** Rarefied community similarity is near-identical on presence (Jaccard 0.536 between habitats against 0.58 within one) and clearly separated on abundance (Bray-Curtis 0.499 against 0.351).

What was rejected

Habitat does not determine species richness. Pooled across all 11 parks the difference appears significant ($p = 4.50e-04$); restricted to the 4 parks surveyed in both habitats it disappears ($p = 0.6855$), and forest wins in exactly 2 of 4. The pooled result was an instance of Simpson's paradox driven by unequal survey effort across parks. Seasonal change could not be assessed at all: visit number and calendar date correlate at $\rho = 0.89$, making a seasonal decline indistinguishable from a repeat-visit decline.

1. Introduction

1.1 Background

Point-count surveys are the standard method for monitoring breeding bird populations. An observer stands at a fixed plot for a set period, records every bird detected by sight or sound, and repeats the visit across a season. The resulting data supports two distinct management questions: which habitats hold the most species, and which habitats hold the species most at risk of decline. These are not the same question, and this analysis finds they have different answers.

The dataset covers the 2018 breeding season across 11 units of the United States National Park Service in the Mid-Atlantic region. Two habitat types were surveyed - forest and grassland - though not, as this report shows, in the same places. Species carrying a Partners in Flight (PIF) Watchlist designation are treated throughout as the at-risk group.

1.2 Objectives

The analysis was structured around six objectives, expressed as fourteen testable questions. Each question is answered in Section 4 with an explicit verdict.

Objective	Questions	Question posed
1. Conservation priority	Q1, Q1b, Q5	Which habitat shelters at-risk species, and is the signal broad or driven by one species?
2. Biodiversity value	Q2, Q11, Q13	Does either habitat support more species, or more species unique to it?
3. Eco-tourism and access	Q3	Which parks and plots offer the richest bird-watching?
4. Habitat dependency	Q4	Which species depend on a single habitat type?
5. Survey optimisation	Q6, Q7, Q8, Q9	When and under what conditions are surveys most productive?
6. Data reliability	Q10, Q12	Is the survey design sound enough to support the above?

Table 1.1 - Objectives and the questions addressing them

1.3 Scope and approach

This analysis deliberately reports negative results alongside positive ones. A question that produced no finding is recorded as firmly as one that did, and Section 9 lists the questions this dataset cannot answer at all. That choice reflects the central methodological lesson of the project: several apparently strong results dissolved once survey effort was accounted for, and a report that presented only surviving findings would misrepresent how the analysis actually went.

1.4 Success criteria

The criteria below were set before the analysis began, so that the project could be judged on whether it answered its questions honestly rather than on whether the answers were interesting.

Criterion	Status	Evidence
Every question receives an explicit verdict	Met	All 14 questions classified as finding, narrowed, rejected or descriptive (Section 4)
No figure is entered by hand anywhere	Met	Dashboard, PDF, Word and ledger all read <code>analysis.run_all()</code>

Criterion	Status	Evidence
Habitat comparisons are confounder-free	Met	Restricted to the 4 shared parks (G2); per-session rates throughout (G1)
Small samples cannot drive a conclusion	Met	30-session floor applied and flagged in every chart (G3)
Negative results are reported as prominently as positive ones	Met	Section 4.3 rejected; Section 9 lists 10 limitations
The cleaning process is auditable	Met	10 decisions logged with row counts in docs/cleaning_log.md
A reader can reproduce every number	Met	Appendix A; each pipeline module runnable independently
Recommendations are specific enough to act on	Met	Section 8; survey-design changes costed in sessions

Table 1.2 - Success criteria and how they were met

2. The dataset

2.1 Source and scale

Source	Two Excel workbooks - one forest, one grassland - supplied as project input
Coverage	11 NPS administrative units, Mid-Atlantic region
Period	2018 breeding season (May to July)
Raw records	8,546 forest rows and 8,531 grassland rows
Cleaned records	15,372 sightings
Survey sessions	1,408
Distinct species	126
Observers	3
Survey plots	609

Seven of the eleven grassland worksheets were empty. Grassland monitoring covered only four parks, and that single structural fact shapes every habitat comparison in this report.

2.2 Survey effort by park and habitat

Park	Forest sessions	Grassland sessions	Total	Both habitats
Antietam National Battlefield	26	252	278	True
Prince William Forest Park	264	0	264	False
Monocacy National Battlefield	30	202	232	True
Chesapeake & Ohio Canal National Historical Park	180	0	180	False
Manassas National Battlefield Park	47	131	178	True
Catoctin Mountain Park	89	0	89	False
National Capital East Parks	58	0	58	False
Harpers Ferry National Historical Park	40	9	49	True
George Washington Memorial Parkway	40	0	40	False
Rock Creek Park	28	0	28	False
Wolf Trap National Park for the Performing Arts	12	0	12	False

Table 2.1 - Survey sessions by park and habitat

The single most important table in this report. 7 of 11 parks were surveyed in one habitat only, and all 7 of those are forest-only. In those parks a forest-versus-grassland comparison is really a park-A-versus-park-B comparison. Only 737 of 1,408 sessions (52.3%) can therefore contribute to the central question.

2.3 Key variables

Variable	Type	Role in the analysis
Admin_Unit_Code	Categorical	Four-letter park code, 11 values
habitat	Categorical	Forest or Grassland (derived from source file)
Plot_Name	Categorical	Survey plot, 609 values
Date / Visit	Date / Integer	Survey date and visit number (1-3)

Variable	Type	Role in the analysis
Observer	Categorical	Surveyor name, 3 values
Start_Time	Time	Session start, used to derive hour and time band
Common_Name / Scientific_Name	Categorical	Species identity
PIF_Watchlist_Status	Boolean	At-risk designation - the conservation variable
Temperature / Humidity	Numeric	Recorded per session
Sky / Wind / Disturbance	Categorical	Ordered condition scales
Flyover_Observed	Boolean	Bird passing overhead rather than using the site
Distance / Interval_Length	Categorical	Detection distance band and time interval

Table 2.2 - Principal variables

A full column-by-column dictionary covering all 49 fields appears in Appendix B.

2.4 Governance, licensing and ethics

Data origin	United States National Park Service inventory and monitoring programme; supplied as project input
Licensing	US federal government works are generally public domain under 17 U.S.C. section 105. No redistribution restriction was supplied with the files, and none is asserted here.
Personal data	The dataset contains 3 named surveyors. These are staff names attached to work records, not personal or sensitive data, but Section 5.1 does report measurable differences in individual performance.
How observer names are handled	Retained, because the observer effect could not be demonstrated without them and it is the project's most important methodological finding. Reported as a property of survey method rather than as an assessment of any individual, and no ranking is framed as a judgement of competence.
Species sensitivity	Plot coordinates are recorded at park level only. No precise nest or roost locations are published, which matters for the at-risk species this report highlights.
Onward use	Figures are reproducible from the pipeline. Anyone citing them should note the single-season scope and the calibration caveat in Section 9.

One ethical decision worth stating plainly. This report identifies which of three named surveyors records fewer species. That is uncomfortable, and it was retained because omitting it would have hidden the largest measured effect in the project and left readers over-trusting the absolute numbers. The finding is presented as evidence about survey method - all three surveyors reach the same conclusions about habitat - and not as a performance review.

3. Methodology

3.1 Cleaning

Ten cleaning decisions were investigated and agreed before any code was written, then applied in a fixed order by `src/clean.py`, which writes an audit trail to `docs/cleaning_log.md` recording row counts before and after each step. Order was not arbitrary: the schema was reconciled while the two habitat files were still separate, and de-duplication ran before any counting, since counting first would have baked a 20% inflation into every grassland figure.

- **Schema reconciliation.** `NPSTaxonCode` renamed to `TaxonCode` after verifying all 88 shared species carry identical codes; `Previously_Obs` dropped as a zero-variance grassland-only column; `Site_Name` retained as nullable, since grassland monitoring genuinely did not record that level.
- **De-duplication.** Exact duplicate rows removed from the grassland file only, where they were an artefact of the source export.
- **Type casting.** Grassland columns had been read as generic objects; cast to integer, float, boolean and datetime so that arithmetic and comparison behave correctly.
- **Value standardisation.** Blank `Sex` values standardised to `Undetermined`; one species recorded under two names unified; `Distance_Display` added while leaving the original `Distance` column untouched.
- **Missing data left in place.** Small scattered gaps were not imputed. Section 5.4 shows why: every gap is explained by a protocol rule rather than by data loss.

3.2 Feature engineering: the survey session

The row-level data records one sighting per row, which makes raw counts a measure of how many birds were seen rather than how good a place is. The analysis therefore aggregates to the **survey session** - one observer, one plot, one morning - producing 1,408 rows. Every rate reported in this document is a column of that table averaged over sessions, never a count taken from the row-level data.

<code>species_per_session</code>	Distinct species recorded - the primary richness measure
<code>sightings_per_session</code>	Total records, used for the at-risk percentage
<code>at_risk_sightings / has_at_risk</code>	Count and presence flag for PIF Watchlist species
<code>time_band</code>	Early (5-6am), Mid (7-8am), Late (9-10am)
<code>temp_band / humidity_band</code>	Binned weather for category comparison
<code>day_of_season</code>	Days since the first survey, for trend testing
<code>is_shared_park</code>	Whether the park was surveyed in both habitats

3.3 Statistical methods

Two tests are used throughout, both implemented in `src/stats_helpers.py` so the project carries no hidden dependency on an external statistics package and every result can be traced to readable source.

- **Mann-Whitney U** for comparing two independent groups. Chosen over a t-test because species-per-session is a bounded count that is not normally distributed, and the test makes no distributional assumption.
- **Spearman's rank correlation (ρ)** for monotonic trends. Section 4.11 documents a case where this statistic is significant and nonetheless misdescribes the relationship, because the underlying curve is not monotonic.

- **Rarefaction** for species-count comparison across unequal sampling effort, using 200 random draws down to the smaller sample size.

Significance is reported at $\alpha = 0.05$ throughout. Exact p-values are given rather than thresholds, so a reader can apply a stricter criterion without recomputation.

3.4 The four guardrails

Four rules were applied to every analysis. They are stated here as method, but each one is also a result: every one changed a conclusion this project would otherwise have published.

ID	Rule	Why it is necessary
G1	Compare per-session rates, never raw totals	Grassland ran 4.2x more sessions than forest. Raw counts measure effort.
G2	Habitat comparisons use the 4 shared parks only	7 parks were surveyed in one habitat only; there, habitat is confounded with park identity.
G3	Treat any group under 30 sessions as unreliable	Small groups swing wildly; the highest single figure in the dataset rests on 12 sessions.
G4	Rarefy before comparing species counts	More sampling finds more species even from an identical community.

Table 3.1 - Analytical guardrails

3.5 Tools and reproducibility

Language	Python 3.13
Core libraries	pandas (data), numpy (numeric), plotly (charts), Streamlit (dashboard), reportlab (this PDF)
Statistics	src/stats_helpers.py - Mann-Whitney U, Spearman rho, no external stats dependency
Pipeline	src/ingest.py, clean.py, features.py, analysis.py - each runnable independently
Single source of numbers	analysis.run_all() - the dashboard, this report and the findings ledger all read from it, so no figure is typed by hand anywhere

3.6 Data flow

Five stages, each a separate module with a single responsibility. Every stage writes its output to disk, so any stage can be re-run and inspected independently rather than the whole thing being one opaque script.

Stage	Module	Input	Output
1. Ingest	src/ingest.py	2 Excel workbooks (15 sheets)	Raw frames, sheet structure report
2. Clean	src/clean.py	Raw frames	data/processed/birds_clean.csv (15,372 rows) + docs/cleaning_log.md
3. Feature build	src/features.py	Cleaned rows	sessions (1,408), species profile (114), park coordinates
4. Analyse	src/analysis.py	Rows, sessions, species	run_all() - a dict of 13 answered questions
5. Present	app/ + src/make_report.py + src/make_ledger.py	run_all() output	Dashboard, PDF, Word, findings ledger

Table 3.2 - Pipeline stages

The critical property is that stage 5 has four consumers and one source. The dashboard, the PDF, the Word document and the findings ledger all read the same `run_all()` dictionary. No presentation layer recomputes a statistic, and none holds a hand-typed number, so the four outputs cannot disagree with each other or with the

analysis.

4. Results

Each question below carries an explicit verdict. **Finding** means the result survived the guardrails and a significance test. **Finding, narrowed** means it survived but in a smaller form than it first appeared. **Rejected** means it did not survive. **Descriptive only** means no test was applied, and the reason is given.

4.1 Q1 - At-risk species by habitat

FINDING, NARROWED

Forest records at-risk species at 3.65% of sightings against 0.59% in grassland - a 6.2x difference, $p = 1.66e-18$, holding in all 4 of 4 shared parks.

Park	Forest	Grassland	Forest higher?
Antietam National Battlefield	2.7	0.42	True
Harpers Ferry National Historical Park	5.92	0	True
Manassas National Battlefield Park	2.37	1.32	True
Monocacy National Battlefield	3.51	0.37	True

Table 4.1 - At-risk sightings as a percentage, by park and habitat (shared parks only)

This is the strongest ecological result in the project. It is computed on shared parks only (G2) and as a per-session rate (G1), and unlike the richness comparison in Section 4.3 it does not collapse when checked park by park: forest is higher in every one of the four. Consistency across parks, rather than the p-value alone, is what distinguishes a real effect from a pooling artefact.

4.1.1 Q1b - Stress-testing the finding

Before accepting the result the at-risk group was decomposed by species. **309 of 378 at-risk sightings (82%) are Wood Thrush alone**, recorded in 10 of the 11 parks. Removing it and repeating the analysis gives a very different picture.

Group	Forest %	Grassland %	Ratio	Parks agreeing
All 8 watchlist species	3.65	0.59	6.2x	4 of 4
Excluding Wood Thrush	0.65	0.31	2.1x	2 of 4

Table 4.2 - The at-risk finding with and without its dominant species

The defensible claim is the narrow one. "Forest shelters at-risk birds" implies a class effect across 8 species. The data supports "forest shelters Wood Thrush": remove that one species and the effect more than halves and stops being consistent across parks. This report uses the narrow phrasing throughout.

4.2 Q5 - Composition of the at-risk group

DESCRIPTIVE ONLY

8 PIF Watchlist species recorded across 378 sightings, of which 81.7% are Wood Thrush.

Species	Forest	Grassland	Total	Parks	% of all at-risk
Wood Thrush	290	19	309	10	81.7
Worm-eating Warbler	31	0	31	5	8.2

Species	Forest	Grassland	Total	Parks	% of all at-risk
Prairie Warbler	7	18	25	3	6.6
Cerulean Warbler	7	0	7	3	1.9
Willow Flycatcher	0	2	2	2	0.5
Kentucky Warbler	1	1	2	2	0.5
Blue-winged Warbler	1	0	1	1	0.3
Red-headed Woodpecker	1	0	1	1	0.3

Table 4.3 - At-risk sightings by species and habitat (all 11 parks)

Reported for all eleven parks rather than the shared four, because this is a question about which species carry the at-risk signal rather than a habitat comparison. The concentration in a single species is the evidence base for narrowing Q1, and is the reason the Overview headline of this project names one bird rather than a category.

4.3 Q2 - Species richness by habitat

REJECTED

Pooled across 11 parks the habitats differ ($p = 4.50e-04$). Within the 4 parks surveyed in both, they do not ($p = 0.6855$).

Comparison	Forest	Grassland	n (F/G)	P	Verdict
Pooled, all 11 parks	8.61	9.15	814 / 594	$p = 4.50e-04$	Appears significant
Within 4 shared parks	9.1	9.15	143 / 594	$p = 0.6855$	Not significant

Table 4.4 - Species per session, pooled versus within shared parks

Park	Forest	Grassland	Forest higher?
Antietam National Battlefield	10.65	9.98	True
Harpers Ferry National Historical Park	8.57	9.22	False
Manassas National Battlefield Park	8.19	8.56	False
Monocacy National Battlefield	9.9	8.49	True

Table 4.5 - Species per session by park and habitat

This is Simpson's paradox, occurring in our own data. Grassland ran 4.2x more sessions than forest, and the two habitats were not surveyed in the same parks. A pooled average therefore partly reflects which parks were surveyed how often rather than habitat quality. Checking within each shared park removes that confound - and the effect vanishes, with forest higher in exactly 2 of 4 parks, which is a coin flip.

The correct conclusion is that richness is the wrong question to ask of habitat in this dataset. Composition - which species are present - does differ between habitats, as Sections 4.1 and 4.5 show. Abundance of species does not.

4.4 Q11 - Species unique to one habitat

FINDING, NARROWED

Raw counts suggest grassland holds 5.1x more exclusive species. Rarefied to equal effort the gap falls to 1.3x.

Measure	Forest-only species	Grassland-only species	Ratio
Raw count	7	36	5.1x
Rarefied (200 draws)	15.5	19.7	1.3x

Table 4.6 - Exclusive species before and after rarefaction

Grassland ran 594 sessions against forest's 143. More sampling finds more species even from an identical community, so the raw comparison largely measures effort. Rarefaction repeatedly resamples the larger group down to the smaller one before counting. A further fragility check: 2 of the forest-only and 15 of the grassland-only species were recorded exactly once, so a single additional sighting would move them into the shared column.

4.5 Q13 - Diversity beyond richness

FINDING

Four independent diversity measures agree there is no habitat difference. Community composition tells a different story: abundance structure differs between habitats (Bray-Curtis 0.499 against 0.351 within a habitat) while species membership barely does.

Counting distinct species is only one way to measure diversity, and the crudest. A session holding ten species - one abundant, nine seen once - scores the same richness as one holding ten evenly abundant species, though the two communities are very different. Three standard indices separate them, each computed per session and then averaged so guardrail G1 holds, inside the shared parks only so G2 holds.

Measure	Forest	Grassland	P	Verdict
Richness	9.105	9.146	p = 0.6855	Not significant
Shannon	2.105	2.084	p = 0.9393	Not significant
Simpson Diversity	0.863	0.854	p = 0.8225	Not significant
Evenness	0.976	0.97	p = 0.1057	Not significant

Table 4.7 - Diversity indices by habitat, per-session means within the shared parks

The null result in Section 4.3 is not an artefact of choosing richness. Shannon weights rare species, Simpson weights dominant ones, and Pielou's evenness removes richness from the calculation entirely. All three reach the same verdict as the species count did: no habitat difference (smallest p = 0.11). Four measures with different sensitivities agreeing is considerably stronger evidence than one measure alone.

4.5.1 Community composition

Diversity indices describe how varied a community is, not whether two communities hold the *same* species. Two further measures compare species lists directly: Jaccard similarity on presence alone, and Bray-Curtis dissimilarity weighted by abundance. Both are strongly effort-sensitive, and forest and grassland session counts inside a park are badly unbalanced, so every figure below is rarefied to the smaller group's session count across 200 draws.

A similarity number means nothing without a yardstick. So the same measures are computed for pairs of parks within a single habitat. If habitat genuinely restructures a community, between-habitat pairs should be less similar than same-habitat pairs. That contrast is the result.

Comparison	Jaccard (presence)	Bray-Curtis (abundance)	Pairs
Between habitats, same park	0.536	0.499	4

Comparison	Jaccard (presence)	Bray-Curtis (abundance)	Pairs
Same habitat, different parks	0.58	0.351	12

Table 4.8 - Rarefied community similarity. Jaccard: higher means more shared species. Bray-Curtis: higher means more different.

Park	Sessions each	Jaccard	Bray curtis	Jaccard raw	Bray curtis raw
ANTI	26	0.613	0.399	0.55	0.812
HAFE	9	0.401	0.679	0.463	0.785
MANA	47	0.588	0.441	0.65	0.576
MONO	30	0.543	0.477	0.495	0.729

Table 4.9 - Forest against grassland within each shared park, rarefied and raw

The two measures disagree, and the disagreement is the finding. Jaccard is essentially flat - 0.536 between habitats against 0.58 within one - so forest and grassland draw on much the same species pool, no more distinct from each other than two parks are. Bray-Curtis separates clearly: 0.499 against 0.351, a gap of 0.148. The habitats hold similar species lists and weight them differently.

Guardrail G4, and how much it mattered here. Before rarefaction the between-habitat Bray-Curtis figure was 0.726 rather than 0.499, which would have overstated the gap against the same-habitat control by roughly 60.5%. Species-list comparisons are among the most effort-sensitive statistics in ecology, and the raw figures are retained in Table 4.9 only to show the size of the correction.

This sharpens the central claim of the report. Section 4.3 established that habitat does not change how many species are present; Section 4.6 will show it does change which ones specialise. This section locates the difference precisely: not in the species list, but in how the community is weighted.

4.6 Q4 - Habitat specialists

FINDING

Of 48 well-sampled species, 17 are grassland specialists, 0 are forest specialists, and 31 are generalists.

Class	Species	Definition
Grassland specialist	17	>80% of sightings in grassland
Generalist	31	Neither habitat exceeds 80%
Forest specialist	0	>80% of sightings in forest

Table 4.10 - Habitat specialisation among 48 well-sampled species

Species	% in grassland	Sightings
Tree Swallow	100	49
European Starling	98.8	82
Grasshopper Sparrow	98.6	351
Barn Swallow	98.5	132
Eastern Kingbird	98	51
Song Sparrow	97.4	228
Red-winged Blackbird	97.2	281
Warbling Vireo	97.1	34

Table 4.11 - The most strongly grassland-loyal species

The zero is the result. Every grassland-associated species seen often enough to classify turned out to be strongly grassland-loyal. Not one well-sampled species showed comparable loyalty to forest - forest birds in this dataset are generalists that also use grassland. Classification uses shared parks only, so a species is not labelled a specialist merely because it was surveyed somewhere with a single habitat type.

This is the finding with the clearest management consequence. Converting grassland removes the only habitat that 17 species reliably use; converting forest removes habitat that no well-sampled species depends on exclusively. Section 4.3 showed richness gives no reason to prefer either habitat - this section shows composition does.

4.7 Q3 - Hotspots: parks and plots

FINDING

Raw species count correlates with survey effort at $\rho = 0.772$ ($p = 0.0053$); the effort-adjusted rate does not ($\rho = 0.227$, $p = 0.5015$).

Park	Sessions run	Species / session	Distinct species	Reliable (30+)?
Antietam National Battlefield	278	10.04	80	True
Chesapeake & Ohio Canal National Historical Park	180	9.95	80	True
National Capital East Parks	58	9.52	66	True
Wolf Trap National Park for the Performing Arts	12	8.83	27	False
Harpers Ferry National Historical Park	49	8.69	54	True
Monocacy National Battlefield	232	8.67	99	True
Manassas National Battlefield Park	178	8.46	80	True
Rock Creek Park	28	8.25	45	False
Prince William Forest Park	264	7.85	54	True
George Washington Memorial Parkway	40	7.45	49	True
Catoctin Mountain Park	89	7.33	46	True

Table 4.12 - Parks ranked by species per session

The two available rankings disagree. Ranked by raw species count Monocacy leads; ranked by species per session it falls to sixth, because it was the third most-visited park. The correlation pair above quantifies why: raw richness is substantially a measure of how often a park was visited, while the rate is independent of effort - which is the property a fair ranking requires.

4.7.1 Plot-level rankings

Plot-level analysis is reported but should not be acted on. All 609 surveyed plots average 8.78 species per session, and no plot was visited more than 3 times. The top-15 leaderboard is simply everything above 13.5 species per session - the right tail of a noisy distribution rather than a set of superior locations. A single unusually good morning is enough to place a plot at the top.

Plot	Park code	Habitat	Species / session	Visits
CHOH-0812	CHOH	Forest	15.5	2
NACE-0477	NACE	Forest	15	2
MONO-0074	MONO	Forest	15	2

Plot	Park code	Habitat	Species / session	Visits
MANA-0240	MANA	Forest	14.5	2
CHOH-0262	CHOH	Forest	14.5	2
ANTI-0163	ANTI	Grassland	14.33	3
NACE-0491	NACE	Forest	14	2
CHOH-1081	CHOH	Forest	14	2
CHOH-0806	CHOH	Forest	14	2
CHOH-0380	CHOH	Forest	14	2

Table 4.13 - Top ten plots by species per session, with visit counts

4.7.2 At-risk presence by park

Park	Sessions run	At-risk sessions	% of sessions	Reliable (30+)?
Catoctin Mountain Park	89	51	57.3	True
Prince William Forest Park	264	107	40.5	True
Rock Creek Park	28	11	39.3	False
Harpers Ferry National Historical Park	49	17	34.7	True
National Capital East Parks	58	11	19	True
Chesapeake & Ohio Canal National Historical Park	180	33	18.3	True
Manassas National Battlefield Park	178	28	15.7	True
George Washington Memorial Parkway	40	5	12.5	True
Antietam National Battlefield	278	22	7.9	True
Monocacy National Battlefield	232	18	7.8	True
Wolf Trap National Park for the Performing Arts	12	0	0	False

Table 4.14 - Percentage of sessions recording an at-risk species

Diversity and conservation point at different parks. Catoctin Mountain Park recorded an at-risk species in 57.3% of sessions - the highest of any park - while ranking #11 of 11 for species per session. A single "best park" recommendation would send birdwatchers and conservation managers to opposite ends of the list.

4.8 Q7 - Time of day

FINDING

Grassland richness falls 17.6% from the early to the late band ($p = 1.75e-05$). Forest shows no effect ($p = 0.5526$).

Habitat	Early (5-6am)	Late (9-10am)	Change	Endpoint p	Trend rho	Trend p	Verdict
Grassland	9.7	8.25	17.6%	$p = 1.75e-05$	-0.166	$p = 4.62e-05$	Significant
Forest	8.65	8.36	3.6%	$p = 0.5526$	-0.037	$p = 0.2938$	Not significant

Table 4.15 - Early versus late richness, tested two ways

Time band	Sessions run Forest	Sessions run Grassland	Species per session Forest	Species per session Grassland
Early (5-6am)	270	179	8.65	9.7
Mid (7-8am)	406	254	8.66	9.32
Late (9-10am)	138	161	8.36	8.25

Table 4.16 - Species per session by time band and habitat

This is the most directly actionable finding in the project, and the one with the strongest evidence: two independent tests - an endpoint comparison and a correlation across the whole morning - agree in both habitats. Two control checks support it. Session length is 10 to 70 minutes with a median of 10 in every band, so the effect is not early surveys running longer. And unlike the monthly comparison in Section 4.9, sessions are distributed across all three bands in both habitats.

Time band	Forest	Grassland
Early (5-6am)	34.1	5.6
Late (9-10am)	35.5	4.3
Mid (7-8am)	30.3	8.7

Table 4.17 - Percentage of sessions recording an at-risk species, by time band

A negative result worth recording: at-risk detection shows no time-of-day pattern in either habitat. The early-morning advantage is about how many species appear, not which ones. Survey scheduling aimed at at-risk species gains nothing from an early start.

4.9 Q6 - Seasonal change

DESCRIPTIVE ONLY

Cannot be assessed. Visit number and day-of-season correlate at $\rho = 0.890$ ($p < 1e-300$), so a seasonal decline is indistinguishable from a repeat-visit decline.

Month	Sessions run Forest	Sessions run Grassland	Species per session Forest	Species per session Grassland
May	212	198	9.22	9.85
June	391	194	8.4	9.15
July	211	202	8.37	8.45

Table 4.18 - Species per session and sessions run, by month

Three confounds sit on the same axis. First, effort is uneven: forest ran 1.85x more sessions in its busiest month than its quietest, while grassland stayed nearly flat at 1.04x. Second, visit number tracks the calendar almost perfectly ($\rho = 0.890$), and richness itself declines with visit number ($\rho = -0.096$, $p = 2.94e-04$) - so the July dip is equally consistent with birds becoming harder to find and with observers recording less on a plot they have already walked twice. Third, third visits happened in grassland only, so late-season figures are a different habitat mix as well as a different time.

No significance test was applied to the monthly figures, and that is deliberate. A p-value computed on confounded groups would lend false authority to a comparison that cannot be made. The numbers are reported descriptively so a reader can see them, labelled so nobody mistakes them for a trend.

4.10 Q9 - Sky, wind and disturbance

FINDING

Serious disturbance costs 36.8% of recorded species (8.93 to 5.65 per session, $p = 2.52e-15$) - the largest effect measured in this project.

Disturbance	Sessions run	Species / session	Reliable (30+)?
Slight effect on count	499	9.38	True
No effect on count	687	8.93	True
Moderate effect on count	160	7.94	True
Serious effect on count	62	5.65	True

Table 4.19 - Species per session by disturbance level

With $n = 687$ undisturbed sessions against $n = 62$ seriously disturbed, this is a 3.28 species gap - larger than the observer effect and 66 times the habitat difference the survey was designed to measure. It is also the only environmental variable that is a property of the site and the schedule rather than of the weather, which makes it the only one management can act on.

Disturbance	Forest	Grassland
Moderate effect on count	7.69	8.23
No effect on count	8.7	9.4
Serious effect on count	5.48	5.79
Slight effect on count	9.14	9.61

Table 4.20 - Disturbance effect within each habitat

An anomaly reported rather than smoothed away. "Slight" disturbance scores *above* "no effect" (9.38 against 8.93), on large samples in both categories, and Table 4.20 shows it replicates independently in forest and in grassland. We have no mechanism to offer. Inventing one would be worse than recording the observation, and the serious-versus-none comparison above is unaffected either way.

Sky	Sessions run	Species / session	Reliable (30+)?
Partly Cloudy	542	9.2	True
Cloudy/Overcast	270	8.76	True
Clear or Few Clouds	496	8.75	True
Fog	61	7.72	True
Mist/Drizzle	39	7.03	True

Table 4.21 - Species per session by sky condition

Wind	Sessions run	Species / session	Reliable (30+)?
Light air movement (1-3 mph) smoke drifts	665	9.22	True
Calm (< 1 mph) smoke rises vertically	398	8.57	True
Light breeze (4-7 mph) wind felt on face	307	8.47	True
Gentle breeze (8-12 mph), leaves in motion	38	7.82	True

Table 4.22 - Species per session by wind condition

Sky and wind show mild, ecologically plausible patterns - fog and drizzle lowest, a stiff breeze costly - but neither was subjected to a significance test as a category comparison, and neither is large enough to justify changing survey practice. They are reported for completeness and marked as untested.

4.11 Q8 - Temperature and humidity

FINDING

Richness peaks at 15-20C and declines at both ends. The relationship is not monotonic, which matters for how the correlation is read.

Temperature band	Sessions run	Species / session	Reliable (30+)?
<15C	50	9.1	True
15-20C	314	9.61	True
20-25C	658	8.91	True
25-30C	313	8.04	True
>30C	73	7.97	True

Table 4.23 - Species per session by temperature band

Humidity band	Sessions run	Species / session	Reliable (30+)?
<40%	12	11.5	False
40-60%	170	8.67	True
60-80%	704	8.68	True
>80%	522	9.03	True

Table 4.24 - Species per session by humidity band

Correlation	Rho	P	Verdict
Forest - Temperature	-0.212	p = 1.02e-09	Significant
Forest - Humidity	0.04	p = 0.2507	Not significant
Grassland - Temperature	-0.191	p = 2.78e-06	Significant
Grassland - Humidity	0.174	p = 2.02e-05	Significant

Table 4.25 - Spearman correlations with weather

A case where a significant statistic misdescribes the data. Spearman's rho assumes a monotonic relationship. Applied to temperature it returns -0.212 for forest and -0.191 for grassland, both significant and both negative - which read alone would tell a manager that the coldest mornings are the most productive. Table 4.23 shows they are not: richness peaks at 15-20C and the coldest band sits below that peak. The defensible claim is that the warm end is worse. Nothing here supports a claim about cold versus mild.

Humidity shows a smaller and less consistent pattern, significant in grassland (p = 2.02e-05) but not in forest (p = 0.2507). The under-40% band records the highest single figure anywhere in this dataset at 11.5 species per session - on 12 sessions, so it falls below the G3 reliability floor and is excluded from every conclusion.

5. Data quality and reliability

Every finding above rests on the survey design. This section audits that design, and contains the result that most nearly invalidated the project.

5.1 Q10 - Observer effects

METHODOLOGICAL

Three surveyors differ by 37.0% in species recorded per session - a gap of 2.69 species against a habitat gap of 0.05. Every pairwise difference is significant.

Observer	Sessions run	Species / session	Spread
Brian Swimelar	461	7.27	2.31
Kimberly Serno	457	9.21	2.74
Elizabeth Oswald	490	9.96	3.26

Table 5.1 - Species per session by observer

Comparison	P	Verdict
Brian Swimelar vs Elizabeth Oswald	p = 2.30e-40	Significant
Brian Swimelar vs Kimberly Serno	p = 3.49e-27	Significant
Elizabeth Oswald vs Kimberly Serno	p = 5.36e-04	Significant

Table 5.2 - Pairwise comparisons between observers

Read that against the ecology before continuing. The habitat difference this survey was built to measure is 0.05 species per session. Which of three people held the clipboard moves the same measure by 2.69 - roughly 54 times more. Had surveyor assignment correlated with habitat, this project would have measured the surveyor and reported it as ecology.

5.1.1 Three checks on whether this invalidates the study

Check 1 - was the rota balanced?

Habitat	Brian Swimelar	Elizabeth Oswald	Kimberly Serno
Forest	0.324	0.357	0.318
Grassland	0.332	0.335	0.333

Table 5.3 - Each observer's share of each habitat's sessions

Yes. Each surveyor covered close to a third of both habitats, with a maximum deviation of 2.4 percentage points (balanced = True). Because assignment is balanced, the observer effect cancels out of any habitat comparison rather than biasing it.

Check 2 - is it the person, or the parks they were sent to?

Park	Brian Swimelar	Elizabeth Oswald	Kimberly Serno
Antietam National Battlefield	8.3	11.56	10.24
Catoctin Mountain Park	6.18	8.45	8.11
Chesapeake & Ohio Canal National Historical Park	8.23	11.28	10.4
Manassas National Battlefield Park	6.9	9.27	9.23
Monocacy National Battlefield	6.74	9.97	9.06

Park	Brian Swimelar	Elizabeth Oswald	Kimberly Serno
Prince William Forest Park	6.84	8.51	8.08

Table 5.4 - Species per session by observer, within each park where all three worked 20+ sessions

The person. Across the 6 parks where all three surveyors worked enough sessions to compare, the ranking is identical in every park - the same surveyor is lowest everywhere and the same one is highest everywhere. The effect cannot be explained by who was sent where.

Check 3 - do the observers disagree about the finding?

Observer	Forest	Grassland
Brian Swimelar	7.45	7.41
Elizabeth Oswald	10.4	10.4
Kimberly Serno	9.12	9.62

Table 5.5 - Species per session by observer and habitat, shared parks only

No. Within the shared parks all three surveyors independently reach the same conclusion - no meaningful richness gap between habitats - even while disagreeing substantially about the absolute numbers. The null result of Section 4.3 replicates three times over.

Verdict: real, large, and harmless here - but only by design. The observer effect does not contaminate any conclusion in this report, because the rota was balanced. That was a design choice, not luck, and it deserves to be recorded as the single most consequential methodological decision in the survey. One practical limit does follow: absolute species-per-session values carry a personal-calibration band of roughly plus or minus 1.3 species, so comparisons within this dataset are sound while the absolute numbers should not be quoted against another study's.

5.2 Q14 - How birds were detected

METHODOLOGICAL

88.2% of all detections were made by ear rather than by eye, and the observer gap of Section 5.1 sits almost entirely on that channel: 2.60 species per session between surveyors on auditory detections against 0.67 on visual.

Section 5.1 established that the observer effect is real, large, and balanced out of the habitat comparison. It did not establish *what* the three surveyors were doing differently. That matters for the recommendations: an effect with no known mechanism can only be managed by rota design, whereas an effect with a located mechanism can be reduced at source. This section locates it.

Detection method	Detections	% of all detections
Singing	9621	62.6
Calling	3941	25.6
Visualization	1808	11.8

Table 5.6 - Detections by identification method

Two of the three methods are auditory. Singing alone accounts for 62.6% of detections and calling for a further 25.6%; only 11.8% of birds were identified by sight. This is normal for a point-count protocol in closed canopy, but it has a consequence that is easy to miss: the survey is, in practice, a hearing test.

Observer	Singing	Calling	Visualization
Brian Swimelar	4.62	2.49	2.16
Elizabeth Oswald	6.46	3.03	2.42
Kimberly Serno	6.07	3.25	1.75

Table 5.7 - Species per session by observer and detection channel

The pattern is not a uniform attentiveness gap. On singing birds the surveyors differ by 1.84 species per session, with Brian Swimelar lowest and Elizabeth Oswald highest. On birds identified by sight the spread narrows to 0.67 - and the ordering changes: Brian Swimelar, who records fewest species overall, is **not** the lowest on the visual channel (Kimberly Serno is). A surveyor who was simply less thorough, or spent less time at the plot, would be lowest on every channel. This one is not.

The observer effect is an ear-training effect, not an effort effect. Roughly 80% of the total between-surveyor gap sits on auditory detections, and the rank order reverses on the visual channel. That is a specific, trainable difference in song and call recognition rather than a general difference in care or diligence. It converts Section 5.1's warning into Recommendation 8.2 - targeted call-recognition calibration - which would not have been justifiable without this breakdown.

Interval	Detections	% of all detections	Cumulative %
0-2.5 min	7755	50.4	50.4
2.5 - 5 min	3149	20.5	70.9
5 - 7.5 min	2386	15.5	86.5
7.5 - 10 min	2082	13.5	100

Table 5.8 - First detections by time interval within the 10-minute count

The 10-minute protocol is recorded in four 2.5-minute intervals, which allows a direct test of whether the count length was adequate. 50.4% of all detections arrive in the first 2.5 minutes and 86.5% within seven and a half; the final interval adds the remaining 13.5%. The accumulation curve is flattening but has not gone flat, so the count length is sufficient for the species that use the site regularly while a longer count would still add occasional records.

The same breakdown by surveyor is reassuring in a different way. The share of detections falling in the first interval ranges from 45.3% to 54.3% across the three surveyors - they accumulate detections at broadly the same rate. Whatever separates them is happening throughout the count, not in a burst at the start or a fade at the end, which is again consistent with recognition skill rather than attention span.

5.3 Q12 - Coverage and effort

DESCRIPTIVE ONLY

4 of 11 parks were surveyed in both habitats, so 52.3% of sessions can address the central question.

Habitat	Count	Mean	Median	Min	Max
Forest	814	10.2	10	10	70
Grassland	594	10.1	10	10	30

Table 5.9 - Session duration by habitat (minutes)

Median session length is 10 minutes in both habitats, so effort per session is directly comparable and no rate in this report requires a duration correction. Forest plots received 2.0 visits on average against 2.96 for grassland -

an imbalance that matters for Section 4.9.

5.4 Protocol adherence and completeness

Sessions on protocol 1,365 of 1,408 (96.9%) ran exactly 10 minutes

Longest session 70 minutes (43 sessions exceeded 10)

Parks below the 30-session floor Rock Creek Park (28), Wolf Trap National Park for the Performing Arts (12)

Column	Missing	% of rows	Explanation
Sub_Unit_Code	14650	95.3	Optional field; most parks have no sub-units
Site_Name	6826	44.4	Recorded for forest plots only, by protocol
Distance	689	4.5	Flyovers have no distance to record
AcceptedTSN	28	0.2	Taxonomic serial number absent for a few records
ID_Method	2	0	Method not recorded on two rows
TaxonCode	2	0	Taxon code not recorded on two rows

Table 5.10 - Missing values and their causes

Missingness is explained by protocol, not by data loss. The clearest case: 689 records carry no Distance value, and exactly those 689 records are flyovers - birds passing overhead rather than using the site, which by definition have no distance to record. The correspondence is one-to-one with no exceptions (verified: True). Missingness that maps perfectly onto a protocol rule is evidence of a well-run survey.

6. Discussion

6.1 Composition matters; abundance does not

The clearest theme across the results is that habitat shapes *which* birds are present far more than *how many*. Section 4.3 found no richness difference once the comparison was made within shared parks. Sections 4.1 and 4.5 found substantial compositional differences in the same data: at-risk sightings concentrated in forest, and every one of the 17 habitat specialists loyal to grassland.

Section 4.5 locates that difference precisely. Rarefied community similarity shows the two habitats draw on much the same species pool - Jaccard 0.536 between habitats against 0.58 between two parks of the same habitat - while their abundance structures separate clearly (Bray-Curtis 0.499 against 0.351). The distinction is not which birds occur but how the community is weighted, which is why specialist counts and at-risk rates detect a habitat effect that species counts cannot.

This has a direct consequence for how such surveys are used. A management decision framed as "which habitat holds more biodiversity" cannot be answered by this data and probably should not be asked - the two habitats hold comparable numbers of species and different sets of them. A decision framed as "which species would we lose" is answerable, and gives an asymmetric answer: converting grassland removes the only habitat 17 species reliably use, while no well-sampled species depends exclusively on forest.

6.2 Effort is the dominant confound

Three separate results in this report were initially misleading for the same reason - unequal survey effort. Pooled richness appeared significant because grassland ran 4.2x more sessions in different parks (Section 4.3). Park rankings by raw species count reflected visit frequency, correlating with effort at $\rho = 0.77$ while the effort-adjusted rate did not (Section 4.7). Exclusive species counts overstated the grassland advantage by roughly four times before rarefaction (Section 4.4).

Each was caught by a different guardrail, but they are the same underlying error: treating a count of observations as a measure of a place. In a dataset where effort varies by a factor of four across the comparison of interest, that error is not a subtle one - it is the default outcome of any un-normalised analysis.

6.3 Measurement error can exceed the signal

The observer effect (2.69 species per session) and the disturbance effect (3.28) both exceed every ecological contrast in this project, including the 0.05-species habitat difference the survey was designed to detect. Neither is an ecological finding; both are properties of how the measurement was taken.

Effect	Gap (species/session)	Significance-tested?
Disturbance (none to serious)	3.28	Yes
Observer (3 surveyors)	2.69	Yes
Temperature (best to worst reliable band)	1.64	Yes
Time of day, grassland	1.45	Yes
Time of day, forest	0.29	Tested, failed
Habitat (within shared parks)	0.05	Tested, failed

Table 6.1 - Every tested effect in the project, in the same units

The practical implication is that survey protocol deserves at least as much design attention as the ecological question. A study that balances its observer rota and controls disturbance can detect a small habitat effect; one that does neither will produce a confident answer about whichever of those two varied most.

6.4 What negative results contributed

Two of the fourteen questions produced no usable answer, and both were informative. The rejected richness comparison (Section 4.3) demonstrated Simpson's paradox in the project's own data and justified restricting every subsequent habitat analysis to shared parks. The unanswerable seasonal question (Section 4.9) identified a design flaw - visit order confounded with calendar date - that a future survey can correct cheaply, and which no amount of post-hoc analysis can repair in this dataset.

7. Insights

Synthesised across questions and ranked by strength of evidence. Each links back to the section that supports it.

#	Insight	Evidence	Section	Strength
1	Forest shelters Wood Thrush specifically, not at-risk birds as a class	3.65% vs 0.59%, $p = 1.66\text{e-}18$, 4/4 parks; but 82% is one species	4.1	Strong, correctly narrowed
2	The surveyor matters more than anything being surveyed	37.0% spread, all pairs significant, consistent in every park, vs a 0.05-species habitat gap	5.1	Strong
3	Grassland holds every habitat specialist; forest holds none	17 vs 0 of 48 well-sampled species	4.5	Strong
4	Disturbance costs more species than any weather variable	36.8% loss, $p = 2.52\text{e-}15$, replicates in both habitats	4.9	Strong
5	Grassland surveys should start early; forest scheduling is free	17.6% gain, two independent tests agree; forest null on both	4.7	Strong, grassland only
6	Habitat does not drive species richness	pooled $p = 4.50\text{e-}04$ collapses to $p = 0.6855$ within shared parks	4.3	Strong negative result
7	"Best park" depends entirely on the question asked	Catoctin Mountain Park ranks #1 for at-risk presence and #11 of 11 for richness	4.6	Solid
8	Habitat restructures abundance, not species membership	rarefied Bray-Curtis 0.499 between habitats vs 0.351 within one; Jaccard essentially flat	4.5	Solid
9	The habitat null result is not a metric artefact	richness, Shannon, Simpson and evenness all agree (smallest $p = 0.11$)	4.5	Strong negative result
10	Raw counts rank effort, not places	raw richness correlates with effort at $\rho = 0.77$; the rate does not ($\rho = 0.23$)	4.6	Solid

Table 7.1 - Insights ranked by strength of evidence

8. Recommendations

8.1 For land management

- **Protect mature forest where Wood Thrush is present, and describe it as such.** The at-risk signal is real and repeats in every shared park, but 82% of it is one species. Framing this as a Wood Thrush measure rather than a general at-risk measure is both more accurate and more defensible if challenged. (Section 4.1)
- **Reduce disturbance during survey windows.** A 36.8% reduction in recorded species is the largest effect measured in this project, and disturbance is the only environmental variable that is a property of the site and schedule rather than the weather. Parks recording moderate or serious disturbance in a large share of sessions are the place to start. (Section 4.10)
- **Do not convert grassland on biodiversity grounds.** Grassland holds all 17 habitat specialists in the dataset while forest holds none, and richness gives no reason to prefer either habitat. A conversion argument built on "more species" has no support in this data. (Sections 4.3 and 4.6)
- **Route at-risk monitoring and diversity visits to different parks.** The park with the highest at-risk detection rate ranks last of 11 for species per session. A single recommendation list would send both audiences to the wrong place. (Section 4.7.2)

8.2 For survey design

These changes would let a future season answer questions this one cannot. None requires additional parks, staff or season length - they redistribute the same effort.

- **Add grassland plots to the 7 forest-only parks.** All 7 single-habitat parks are forest-only, holding 671 sessions that currently cannot contribute to the central question. Pairing habitats within those parks would take usable data from 52.3% toward 100% - roughly 91% more analysable sample. This is the single highest-value change available. (Section 2.2)
- **Document the balanced observer rota as a protocol requirement.** The 37.0% observer spread is harmless here purely because assignment was balanced across habitats and parks. That currently exists as practice rather than as a written rule, and it is the reason this study's conclusions hold. (Section 5.1)
- **Calibrate surveyors on song and call recognition specifically.** 88.2% of detections are auditory, and 2.60 of the 2.69-species observer gap sits on that channel against 0.67 on the visual one - with the ranking reversing between the two. This is a trainable recognition difference, not a difference in diligence, so a short pre-season call-identification calibration targets the actual mechanism. (Section 5.2)
- **Decouple visit number from calendar date.** The two correlate at $\rho = 0.89$, which makes seasonal analysis impossible. Rotating visit order across plots - surveying some plots in reverse order - would unlock a whole objective at no additional cost. (Section 4.9)
- **Equalise visits per plot across habitats.** Forest plots received 2.0 visits on average against grassland's 2.96, and third visits happened in grassland only. (Section 5.3)
- **Lift the smallest parks over the 30-session floor.** Only two parks fall below it: Rock Creek Park needs 2 more; Wolf Trap National Park for the Performing Arts needs 18 more. That is 20 extra sessions to make two more parks reportable - the cheapest improvement on this list. (Section 5.4)
- **Increase visits per plot beyond three.** No plot exceeded 3 visits, which makes plot-level ranking impossible. Plot-scale recommendations require substantially more repeat sampling. (Section 4.7.1)

9. Limitations

Presented as a numbered section rather than a footnote. Knowing which questions this dataset cannot answer is part of the result, and several of the entries below are the direct output of the guardrails described in Section 3.4.

Limitation	Explanation	Section
Seasonal trend	Visit number and day-of-season correlate at $\rho = 0.89$, and third visits were grassland-only. A seasonal decline cannot be separated from a repeat-visit decline.	4.9
Strength of the habitat null result	Section 4.3 rejects a habitat effect on richness, but does so within 4 parks on 143 forest sessions. That is enough to reject a strong claim, not enough to demonstrate equivalence across the park system.	4.3
Individual plot quality	No plot was visited more than 3 times. The plot leaderboard is the right tail of a noisy distribution, not a set of superior locations.	4.6.1
Cold versus mild mornings	The temperature relationship is hump-shaped, so the significant negative correlation misdescribes the cold end.	4.10
At-risk birds and survey timing	No time-of-day effect on at-risk detection in either habitat. The morning advantage concerns how many species appear, not which.	4.7
Mechanism for the disturbance anomaly	Unknown. Large samples on both sides, replicated in both habitats, and no mechanism available. Reported as observed.	4.9
Absolute species-per-session values	Carry a personal-calibration band of roughly plus or minus 1.3 species from the observer effect. Within-dataset comparisons are sound; the absolute figures should not be quoted against another study's.	5.1
The 7 single-habitat parks	They contain no within-park habitat comparison and contribute nothing to the central question.	2.2
Sky and wind effects	Reported descriptively; never subjected to a significance test as a category comparison, so the apparent ordering is unverified.	4.9
Single season	All data comes from one breeding season. Between-year variation is unmeasured, so no trend over time can be inferred.	2.1

Table 9.1 - Limitations and their sources

10. Conclusion

Thirteen questions were asked of 15,372 bird sightings across 1,408 survey sessions. Five produced findings that survived scrutiny, one was rejected, and the remainder were descriptive or methodological. The surviving findings are consistent with each other and point in the same direction: habitat in this dataset determines *which* birds are present rather than *how many*.

The strongest single result is not ecological. The difference between three surveyors (2.69 species per session) and the effect of survey disturbance (3.28) both exceed the habitat difference the study was designed to detect (0.05). The analysis holds because the observer rota was balanced across habitats and parks - a design decision that, on the evidence assembled here, did more to protect this study's validity than any analytical

choice made afterwards.

Three of the thirteen questions were answered by rejecting them, and those rejections were as informative as the confirmations. A pooled habitat comparison, an un-normalised park ranking and a raw exclusive-species count would each have produced a confident, publishable and wrong answer. They were caught by four rules applied consistently rather than by any single test, which suggests the rules themselves are the transferable output of this project.

For management, the actionable conclusions are narrow but firm: protect forest where Wood Thrush is present, reduce survey-window disturbance, retain grassland because it holds every habitat specialist in the dataset, and survey grassland early. For the next survey season, the highest-value change is structural rather than analytical - adding grassland plots to the 7 forest-only parks would increase the analysable sample by roughly 91% without a single additional park, staff member or survey day.

Appendix A - Reproducibility

Every figure in this report is generated by the analysis pipeline at build time. No number is typed by hand, in this document or in the dashboard, so the two cannot disagree.

Module	Responsibility
src/ingest.py	Reads the two source workbooks, reports sheet-level structure
src/clean.py	Applies the ten cleaning decisions; writes docs/cleaning_log.md
src/features.py	Builds the session table, species profile and park coordinates
src/stats_helpers.py	Mann-Whitney U, Spearman rho, rank utilities
src/analysis.py	Answers Q1-Q12; run_all() is the single source of every figure
src/report_content.py	This report, as structured content
src/make_report.py	Renders that content to PDF and Word
src/make_ledger.py	Renders the findings ledger to markdown
app/	Streamlit dashboard - reads the same run_all() output

Table A.1 - Pipeline modules

To reproduce this report	python src/make_report.py - writes docs/ from the current data
To reproduce the analysis	python src/analysis.py - prints every headline figure
To reproduce the cleaning audit	python src/clean.py - rewrites docs/cleaning_log.md
Generated	31 August 2026

Appendix B - Data dictionary

All 49 fields in the cleaned dataset, split into those that came from the source workbooks and those the pipeline derived. Types and distinct-value counts are read from the data itself at build time, so this table cannot drift out of step with the file it describes.

B.1 Fields from the source workbooks

Field	Type	Distinct	Missing	Description
Admin_Unit_Code	str	11	0	Four-letter park code
Sub_Unit_Code	str	8	14,650	Sub-unit within a park; optional
Site_Name	str	70	6,826	Survey site between park and plot; forest only
Plot_Name	str	609	0	Individual survey plot
Location_Type	str	2	0	Forest or Grassland, as recorded in the source
Year	int64	1	0	Survey year
Date	datetime64[us]	64	0	Survey date
Start_Time	str	318	0	Session start time
End_Time	str	318	0	Session end time
Observer	str	3	0	Surveyor who recorded the session
Visit	int64	3	0	Visit number to that plot (1-3)
Interval_Length	str	4	0	Time interval within the count when detected
ID_Method	str	3	2	How the bird was identified (singing, calling, visual)
Distance	str	2	689	Detection distance band; blank for flyovers
Flyover_Observed	bool	2	0	Bird passing overhead rather than using the site
Sex	str	3	0	Recorded sex, or Undetermined
Common_Name	str	126	0	Species common name
Scientific_Name	str	126	0	Species binomial - the identity key used here
AcceptedTSN	float64	125	28	Taxonomic Serial Number
TaxonCode	float64	125	2	NPS taxon code

Field	Type	Distinct	Missing	Description
AOU_Code	str	126	0	American Ornithological Union four-letter code
PIF_Watchlist_Status	bool	2	0	Partners in Flight at-risk designation
Regional_Stewardship_Status	bool	2	0	Regional stewardship designation
Temperature	float64	210	0	Air temperature at the session, degrees C
Humidity	float64	432	0	Relative humidity at the session, %
Sky	str	5	0	Sky condition category
Wind	str	4	0	Beaufort-style wind category
Disturbance	str	4	0	How much disturbance affected the count
Initial_Three_Min_Cnt	bool	2	0	Detected in the first three minutes

Table B.1 - 29 source fields

B.2 Fields derived by the pipeline

Field	Type	Distinct	Missing	Description
Distance_Display	str	3	0	Cleaned display form of Distance
session_id	str	1,408	0	Plot + date + visit - the unit of every rate
habitat	str	2	0	Forest or Grassland, normalised
park_name	str	11	0	Full park name
is_shared_park	bool	2	0	Park surveyed in both habitats - drives G2
year	int64	1	0	Year, derived from Date
quarter	str	2	0	Calendar quarter
month	str	3	0	Year-month
month_name	str	3	0	Month name
week	int64	11	0	ISO week number
day_of_season	int64	64	0	Days since the first survey of the season
start_hour	int64	6	0	Hour the session began
time_band	str	3	0	Early (5-6am), Mid (7-8am), Late (9-10am)
session_duration_min	float64	6	0	Session length in minutes

Field	Type	Distinct	Missing	Description
is_at_risk	bool	2	0	Boolean form of PIF_Watchlist_Status
is_stewardship	bool	2	0	Boolean form of Regional_Stewardship_Status
is_flyover	bool	2	0	Boolean form of Flyover_Observed
in_first_three_min	bool	2	0	Boolean form of Initial_Three_Min_Cnt
temp_band	str	5	0	Binned temperature
humidity_band	str	4	0	Binned humidity

Table B.2 - 20 derived fields, added by src/features.py

B.3 The session table

Analysis operates on the session table rather than on individual sightings. It has 1,408 rows and carries the columns below in addition to the session-level attributes above.

Field	Type	Description
species_per_session	int64	Distinct species recorded - primary richness measure
sightings_per_session	int64	Total records in the session
at_risk_sightings	int64	Records of PIF Watchlist species
flyovers	int64	Records flagged as flyovers
pct_at_risk_per_session	float64	At-risk records as a % of the session
has_at_risk	bool	Whether any at-risk species was recorded

Table B.3 - Session-level metrics